

Jiayi Zhang

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Education

University of Nottingham Ningbo China (UNNC)

BSc (HONS) MATHEMATICS WITH APPLIED MATHEMATICS

- GPA - 3.94/4.00

Ningbo China

Sep 2024 - Jun 2028 (Expected)

Research Interests

My research focuses on **foundation-model-based robot learning** for reliable, **long-horizon manipulation**. I study how foundation models, structured task-and-motion reasoning, and **geometry-aware visuomotor policies** can be integrated through effective **action representations**, reusable skill abstractions, and policy-controller co-design. My goal is to enable robots to ground semantic goals into closed-loop physical behaviors, adapt and recover under uncertainty, and generalize to complex human-centered tasks such as **tabletop Mahjong**.

Experience

Tsinghua University (THU)

RESEARCH ASSIST, ADVISED BY [PROF. MENGDI XU](#)

- **Research Topics:** Manipulation, In-Context Learning, Robot learning

Beijing, China

July 2026- Now

Shanghai Jiao Tong University (SJTU)

RESEARCH ASSIST AT [IWIN-FINS LAB](#), ADVISED BY [PROF. JIANPING HE](#)

- **Research Topics:** Robot learning, Control, Manipulation.

Robot arm: Project leader

D.I.V.E FORUM: Organizer

Underwater 3D reconstruction: Project leader

WBCD 2nd dual arm robot: member

Shanghai, China

June 2025 - June 2026

The University of Hong Kong (HKU)

RESEARCH INTERN AT [IoT LAB](#), ADVISED BY [PROF. EDITH C.H. NGAI](#)

- **Research Topics:** federated learning, fairness, spatio-temporal graphs.

Hong Kong SAR, China

March 2025 – August 2025

Publications

PREPRINTS

[P2] RoboCoach: World Models as Active Coaches for Compositional Robot Skills

Jiajun Liu*, Yifan Chen*, Yichao Liu*, [Jiayi Zhang](#), Ruoqu Chen, Shaoxuan Xie, Guocai Yao, Mengdi Xu, Sen Cui, Changshui Zhang

Under review, 2026 [\[Project\]](#)

[P1] F²STNet: Fair and Federated Spectral-Temporal Modeling for Graph Forecasting

Jiayi Zhang*, Jinfeng Xu*, Hewei Wang*, Siyuan Cen, Haidong Huang, Yiyao Zhan, Zheyu Chen, Jinjiang You, Ai Jian, Edith C. H. Ngai

Under review, 2025

CONFERENCE PROCEEDINGS

[C5] CoRDE: Concept-Prior Routed Diffusion Experts for Structural Generalization in Robot Manipulation

Haidong Huang, Xixin Zhao, Yaohua Zhou, Jiayu Song, [Jiayi Zhang](#), Jun Ma, Haiyue Zhu, Xiaocong Li

IROS, 2026 [\[Paper\]](#)

[C4] Multi-modal Dynamic Proxy Learning for Personalized Multiple Clustering

Jinfeng Xu, Zheyu Chen, Shuo Yang, Jinze Li, Ziyue Peng, Zewei Liu, Hewei Wang, [Jiayi Zhang](#), Edith C. H. Ngai

AAAI, 2026 [\[Paper\]](#)

[C3] Multi-Cali Anything: Dense Feature Multi-Frame Structure-from-Motion for Large-Scale Camera Array Calibration

Jinjiang You*, Hewei Wang*, Yijie Li, Mingxiao Huo, LVT Ha, Mingyuan Ma, Jinfeng Xu, [Jiayi Zhang](#), Puzhen Wu, Shubham Garg, Wei Pu

IROS, 2025 (Oral) [\[Paper\]](#)

[C2] Spec-LLAVA: Accelerating Vision-Language Models with Dynamic Tree-Based Speculative Decoding

Mingxiao Huo*, [Jiayi Zhang](#)*, Hewei Wang*, Jinfeng Xu, Zheyu Chen, Huilin Tai, Ian Yijun Chen

ICML Workshop, 2025 [\[Paper\]](#)

[C1] Segregation and Context Aggregation Network for Real-Time Cloud Segmentation

Yijie Li*, Hewei Wang*, [Jiayi Zhang](#), Jinjiang You, Jinfeng Xu, Puzhen Wu, Yunzhong Xiao, Soumyabrata Dev

ICLR Workshop, 2025 [\[Paper\]](#)

JOURNAL ARTICLES

[J2] TrafficKAN-GCN: Graph Convolutional-based Kolmogorov-Arnold Network for Traffic Flow Optimization

Jiayi Zhang*, Yiming Zhang*, Yuan Zheng, Yuchen Wang, Jinjiang You, Yuchen Xu, Wenxing Jiang, Soumyabrata Dev
arXiv preprint, 2025 [\[Paper\]](#)

[J1] RAINER: A Robust Ensemble Learning Grid Search-Tuned Framework for Rainfall Patterns Prediction

Zhenqi Li*, Junhao Zhong*, Hewei Wang, Jinfeng Xu, Yijie Li, Jinjiang You, [Jiayi Zhang](#), Runzhi Wu, Soumyabrata Dev
arXiv preprint, 2025 [\[Paper\]](#)

Honors & Awards

2026 **Outstanding student (Top 1% student)**, UNNC Award for Outstanding Students (UG)

Ningbo China

2025 **Provost's Scholarship (Top 5% student)**, University Academic Excellence Scholarship

Ningbo China

2025 **Third Prize & Best Researcher Award**, 9th Golden Interns Award

Ningbo China

2025 **Honorable Mention**, Interdisciplinary Contest in Modeling 2025

Online

Academic Services

Reviewer NeurIPS, AAAI, AISTATS

2025–Present

Leadership/Mentorship/Social

Semester Peer Mentor

UNIVERSITY OF NOTTINGHAM NINGBO CHINA (UNNC)

- Provided academic guidance and practical resources for freshmen.

Ningbo China

Sep. 2025–Jun. 2026

Skills

Programming Python, C++, $\text{\LaTeX}$, MATLAB, HTML, CSS, R

Frameworks PyTorch, Tensorflow, NumPy, Flask, MySQL, Git, Anaconda, OpenCV, ROS

Models CNN, ResNet, Faster R-CNN, YOLO, LSTM, Transformer, BERT, ViT, GANs, VAE, Stable Diffusion

Platforms Unitree Z1 Pro, Franka Research 3, AgileX Piper

Additional

Language Chinese (native), English (working proficiency)

Interests Running, hiking, mahjong, chessing, pickleball, roguelike and strategy games